

**ONLINE RESULT PUBLISHING  
WEBSITE AI23431- WEBTECHNOLOGY  
AND MOBILE APPLICATION  
MINI-PROJECT REPORT**

**SUBMITTED BY**

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**(241801131)**

*in partial fulfillment for the award of the degree of*

**BACHELOR OF TECHNOLOGY**

*in*

**ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**



**DEPARTMENT OF ARTIFICIAL INTELLIGENCE  
AND DATA SCIENCE**

**RAJALAKSHMI ENGINEERING COLLEGE**

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**APRIL 2026**

# **RAJALAKSHMI ENGINEERING COLLEGE**

**(an Autonomous Institution Affiliated to Anna University, Chennai)**

## **BONAFIDE CERTIFICATE**

Certified that this Project Thesis titled “**ONLINE RESULT PUBLISHING WEBSITE**” is the Bonafide work of V S LAKSHANIKA (2116241801131) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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## **DEPARTMENT VISION**

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

## **DEPARTMENT MISSION**

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

## **PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)**

**PEO 1:** To equip students with an essential background in computer science, basic electronics and applied mathematics.

**PEO 2:** To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

**PEO 3:** To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

**PEO 4:** To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

## **PROGRAMME OUTCOMES (POs)**

**PO 1: Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

**PO 2: Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

**PO 3: Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

**PO 4: Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

**PO 5: Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

**PO 6: The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

**PO 7: Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**PO 8: Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

**PO 9: Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

**PO 10: Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

**PO 11: Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

**PO 12: Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

## **PROGRAM SPECIFIC OUTCOMES (PSOs)**

A graduate of the Artificial Intelligence and Data Science Program will demonstrate

**PSO 1: Foundation Skills:** Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

**PSO 2: Problem-Solving Skills:** Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

**PSO 3: Successful Progression:** Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

## **COURSE OBJECTIVE**

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

## **COURSE OUTCOME**

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other



## CO-PO-PSO Mapping

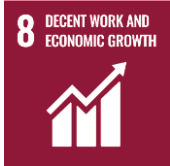
| CO      | PO 1 | P<br>O<br>2 | P<br>O<br>3 | P<br>O<br>4 | P<br>O<br>5 | P<br>O<br>6 | P<br>O<br>7 | P<br>O<br>8 | P<br>O<br>9 | P<br>O<br>10 | P<br>O<br>11 | P<br>O<br>12 | PS<br>O<br>1 | PS<br>O<br>2 | PS<br>O<br>3 |
|---------|------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|--------------|--------------|--------------|--------------|--------------|--------------|
| CO<br>1 | 3    | 3           | 3           | 3           | 3           | 2           | 2           | 2           | 3           | 2            | 3            | 3            | 3            | 3            | 3            |
| CO<br>2 | 3    | 3           | 3           | 3           | 3           | 2           | -           | -           | 2           | 2            | 2            | 3            | 3            | 2            | 2            |
| CO<br>3 | 3    | 3           | 3           | 2           | 3           | 1           | 1           | 2           | 3           | 3            | 3            | 3            | 3            | 3            | 3            |
| CO<br>4 | 3    | 3           | 3           | 3           | 3           | 2           | 1           | 2           | 3           | 2            | 2            | 3            | 3            | 3            | 3            |
| CO<br>5 | 1    | 1           | 1           | 1           | 1           | -           | -           | -           | 3           | 3            | 3            | 3            | 1            | -            | 2            |

**Note:** Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low)    2: Moderate (Medium)    3:

Substantial (High) No correlation: “-”

## SUSTAINABLE DEVELOPMENT GOALS

| SDG                                                                                                      | GOAL                                                   | JUSTIFICATION                                                                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>SDG-4</b></p>    | <p><b>Quality Education</b></p>                        | <p><b>Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.</b></p>                                                                          |
|  <p><b>SDG-8</b></p>   | <p><b>Decent Work and Economic Growth</b></p>          | <p><b>Necessary for social stability, shared prosperity, and combating poverty.</b></p>                                                                                                             |
|  <p><b>SDG-9</b></p>  | <p><b>Industry, Innovation, and Infrastructure</b></p> | <p><b>Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.</b></p> |
|  <p><b>SDG-17</b></p> | <p><b>Partnerships for the Goals</b></p>               | <p><b>Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.</b></p>      |

## ABSTRACT

This project is an **online result publishing website** designed to simplify the process of publishing, viewing, and managing academic results. It provides a convenient platform where students can check their results, administrators can publish exam scores, and faculty can manage student records efficiently.

The application allows administrators to create accounts, log in, and manage result entries. Administrators can add new results with details such as student ID, subject codes, marks, grades, and semester information, as well as view complete academic records. A search feature is also included, enabling students to retrieve their results by entering their roll number or registration number.

The system is developed using **Java, JSP, and MySQL**, and follows a structured approach to keep different parts of the application organized. Basic security features such as user authentication, role-based access control, and safe database operations are implemented to ensure reliability and data integrity.

Overall, the project demonstrates how web technologies can be used to build a practical and user-friendly platform for academic result management and publication.

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## LIST OF ABBREVIATIONS

| Abbreviation | Full Form                      |
|--------------|--------------------------------|
| HTML         | HyperText Markup Language      |
| CSS          | Cascading Style Sheets         |
| JSP          | Java Server Pages              |
| JDBC         | Java Database Connectivity     |
| SQL          | Structured Query Language      |
| DBMS         | Database Management System     |
| UI           | User Interface                 |
| UX           | User Experience                |
| HTTP         | HyperText Transfer Protocol    |
| CGPA         | Cumulative Grade Point Average |
| GPA          | Grade Point Average            |
| PDF          | Portable Document Format       |

# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 General**

In today's digital era, the internet has become a primary medium for accessing information and services. The education sector is no exception, as more institutions prefer to publish examination results online rather than relying on traditional methods such as notice boards or printed mark sheets. Online platforms provide students with quick access to their academic results, making the process of result checking more efficient, transparent, and convenient.

This project focuses on the development of an online result publishing website that allows students to check their examination results, view detailed mark sheets, and track their academic performance. The system is designed to be user-friendly and accessible, enabling administrators to publish results efficiently and students to retrieve their scores instantly by entering their roll number or registration number.

### **1.2 Need for Online Result Publishing System**

Traditional methods of result publication are often time-consuming, less transparent, and prone to errors. Students usually have to wait for physical notice boards to be updated or collect printed mark sheets from the institution, which can lead to delays and inconvenience. Similarly, administrators may find it difficult to manage and distribute results to a large number of students efficiently.

An online result publishing system addresses these challenges by providing a centralized platform where results can be uploaded once and accessed by multiple users simultaneously. It reduces paperwork, improves accessibility, and allows students to check their results from anywhere at any time. Additionally, it ensures that information is updated in real time, making the process more reliable, transparent, and efficient.

### 1.3 Scope of the System

The system is designed to serve as a comprehensive platform for result publication and management. It allows administrators to log in, add student results with relevant details such as roll number, name, subject codes, marks obtained, grades, and semester information. The system also provides search functionality to help students find their results quickly by entering their unique identification number.

Furthermore, students can view detailed mark sheets including subject-wise scores, total marks, percentage, and grade. The system is intended for educational institutions such as schools, colleges, and universities, and can be extended in the future with additional features such as grade analytics, result comparison, PDF download, SMS notifications, and mobile application support.

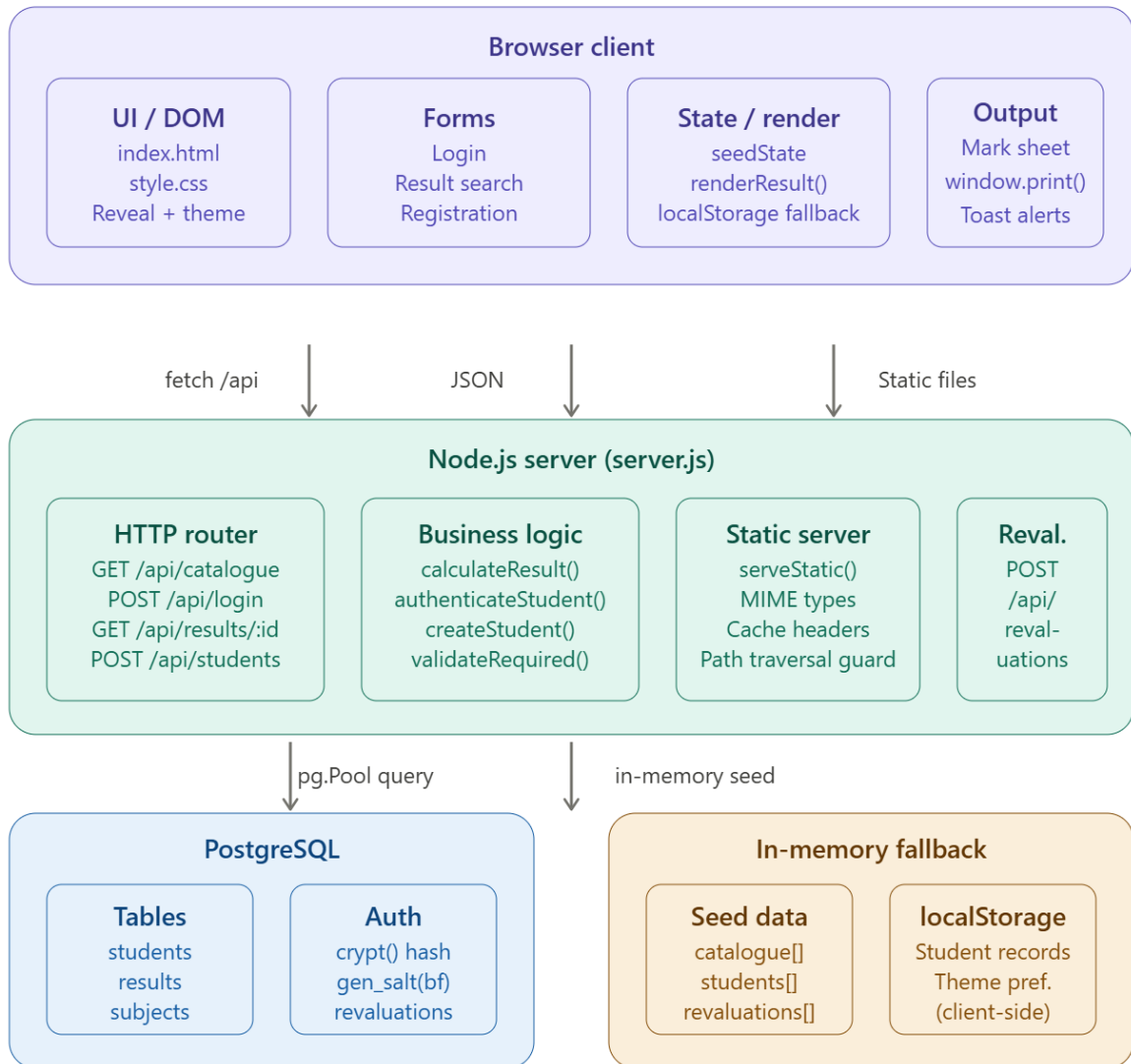
### 1.4 Organization of the Report

This report is organized into several chapters that describe the development and implementation of the online result publishing system.

- **Chapter 1** provides the introduction to the project, including the general overview, need for the system, and scope of the project.
- **Chapter 2** presents the literature survey, which discusses existing result management platforms and related work.
- **Chapter 3** explains the problem formulation and objectives of the proposed system.
- **Chapter 4** describes the system design, including system architecture, database design, and system requirements.
- **Chapter 5** discusses the implementation of the system and explains the different modules.
- **Chapter 6** presents the results and discussion, including system outputs and performance evaluation.
- **Chapter 7** concludes the project and suggests possible future enhancements.



## System architecture diagram:



## **CHAPTER 2**

### **LITERATURE SURVEY**

#### **2.1 Introduction**

The literature survey explores existing systems and approaches used in online result publishing and academic management platforms. With the rapid growth of internet usage and digital literacy, educational institutions have increasingly adopted online solutions for result publication. This section examines how current platforms function, the technologies they use, and the challenges they face. Understanding these aspects helps in designing a system that is both efficient and user-friendly while addressing the limitations of existing solutions.

#### **2.2 Existing Result Management Systems**

Modern result management platforms provide students and administrators with the ability to publish, view, and manage examination results digitally. These systems typically offer features such as roll number-based search, subject-wise mark display, grade calculation, semester-wise result tracking, and downloadable mark sheets. Administrators can upload results in bulk or individually and manage student records through centralized dashboards.

While these platforms are effective and widely used, they often come with certain limitations. Many systems are designed for large-scale institutional use, making them complex and sometimes difficult for smaller institutions to implement. In addition, some platforms lack proper security measures, making student data vulnerable to unauthorized access. Excessive advertisements, slow loading times, or incomplete result information can also affect the overall user experience and trust in the system.

#### **2.3 Related Work**

Previous studies and projects in the domain of web-based result management systems have focused on improving usability, security, efficiency, and accessibility. Researchers have explored the use of dynamic web technologies to create interactive user interfaces and faster data retrieval systems. Database-driven applications are commonly used to manage large volumes of student and result data, ensuring that users can access updated and accurate information instantly.

Security and user authentication are also key areas of focus in related work, with systems implementing role-based access control, encrypted passwords, and session management to protect sensitive academic data. Some solutions incorporate advanced features such as grade analytics, performance graphs, result comparison across semesters, automated notifications via email or SMS, and integration with student information systems. However, these features often increase system complexity and may not be necessary for all institutions, especially smaller educational bodies with limited technical resources.

## **2.4 Summary**

The analysis of existing platforms and related work highlights the importance of providing a balanced system that combines functionality with simplicity. While many advanced systems offer a wide range of features, there is still a need for applications that are easy to use, secure, efficient, and focused on essential functionalities.

Based on this study, the proposed system aims to deliver a streamlined result publishing platform that enables administrators to publish results and students to check their scores with ease. The focus is on creating a reliable, transparent, and user-friendly application that meets core requirements without unnecessary complexity.

## **CHAPTER 3**

### **PROBLEM FORMULATION AND OBJECTIVES**

#### **3.1 Problem Statement**

In the current educational environment, examination result publication is often carried out through traditional methods such as physical notice boards, printed mark sheets, or manual distribution. These approaches can be time-consuming, inefficient, lack transparency, and are prone to errors or delays. Students may struggle to access their results promptly, while administrators face difficulties in managing and distributing results to a large number of students efficiently.

Existing online platforms, although feature-rich, are often complex and may include unnecessary functionalities for basic institutional needs. They can also lack proper security measures, putting sensitive student data at risk. Additionally, many systems do not provide a simple and centralized interface for result publication and retrieval.

Therefore, there is a need for a streamlined and user-friendly web-based system that allows administrators to easily publish examination results and enables students to check their scores quickly and securely using their roll number or registration number.

#### **3.2 Objectives of the System**

The main objective of the system is to develop a web-based application that simplifies the process of result publication and retrieval. The specific objectives are as follows:

- To design and develop a user-friendly interface for result publication and viewing
- To enable students to search and view their results using roll number or registration number
- To provide functionality for administrators to add and manage student result records
- To allow students to view detailed mark sheets including subject-wise marks and grades
- To implement a secure login and authentication system for administrators using session management

- To ensure safe and efficient database operations using proper connectivity techniques
- To calculate and display total marks, percentage, and overall grade automatically
- To create a structured and maintainable system using appropriate design patterns

## **CHAPTER 4**

### **SYSTEM DESIGN**

#### **4.1 Introduction**

System design is an important phase in software development where the overall structure and components of the system are planned. It defines how different modules interact with each other and how data flows within the system. A well-designed system improves efficiency, reliability, and ease of maintenance.

The online result publishing system is designed as a web-based application that allows administrators to publish results and students to view their academic scores. It provides features such as admin authentication, result entry, student result search, and detailed mark sheet display. The system follows a client-server architecture where users interact through web pages, while the backend processes requests and communicates with the database.

#### **4.2 System Architecture**

The system follows a three-layer architecture consisting of the presentation layer, application layer, and data layer.

##### **Presentation Layer**

This layer represents the user interface where users interact with the system. It includes web pages such as the homepage, student search page, result display page, admin login page, and result entry page. These pages are developed using JSP, HTML, and CSS, and are responsible for displaying data to users and collecting input.

##### **Application Layer**

This layer handles the core logic of the system. It processes user requests, manages sessions, performs validation, calculates grades and percentages, and controls the flow of data between the presentation layer and the database.

The backend is implemented using **Java Servlets**, where each servlet is responsible for a specific functionality:

- **AdminLoginServlet** → Handles administrator authentication
- **AddResultServlet** → Adds new student result records
- **SearchResultServlet** → Searches results by roll number
- **ViewResultServlet** → Displays detailed mark sheet
- **UpdateResultServlet** → Updates existing result records
- **DeleteResultServlet** → Deletes result entries
- **LogoutServlet** → Manages admin logout

This layer ensures proper execution of all operations and maintains system functionality.

### **Data Layer**

This layer is responsible for storing and managing all application data. The system uses a MySQL database connected through JDBC for data operations.

The database consists of three main tables:

- **admin** → Stores administrator credentials (admin\_id, username, password)
- **students** → Stores student information (student\_id, roll\_number, name, class, section)
- **results** → Stores examination results (result\_id, student\_id, subject, marks, grade, semester, exam\_type)

### **4.3 Database Design**

The database is designed to efficiently store and manage data related to users, properties, and inquiries. Each table is structured with appropriate fields and relationships.

- The **admin** table stores login credentials for authorized users
- The **students** table stores registered student information
- The **results** table stores subject-wise marks and is linked to students through a foreign key

Relationships between tables ensure data consistency and integrity. For example, each result entry is associated with a specific student, and multiple results can exist for the same student across different subjects and semesters.

**Key Fields in Results Table:**

- result\_id (Primary Key)
- student\_id (Foreign Key referencing students table)
- roll\_number
- subject\_name
- marks\_obtained
- max\_marks
- grade
- semester



## 4.4 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

### 4.4.1 Hardware Requirements

The system does not require high-end hardware and can run on standard computers.

- **Processor:** Intel Core i3 or higher
- **RAM:** Minimum 4 GB
- **Storage:** 500 GB Hard Disk or SSD
- **Input Devices:** Keyboard and Mouse
- **Internet Connection:** Required

### 4.2.1 Software Requirements

The system is developed using commonly available tools and technologies.

- 5 **Operating System:** Windows 10 or later
- 6 **Frontend Technologies:** HTML, CSS, JSP
- 7 **Backend Technology:** Java Servlets
- 8 **Database:** MySQL
- 9 **Database Connectivity:** JDBC
- 10 **Web Server:** Apache Tomcat 9.0
- 11 **Development Environment:** Eclipse IDE for Enterprise Java
- 12 **Web Browser:** Google Chrome or any modern browser

## CHAPTER 5

### SYSTEM IMPLEMENTATION

#### 5.1 System Methodology

The system is developed using a structured approach based on the Model-View-Controller (MVC) architecture. This methodology helps in separating the user interface, business logic, and data handling, making the system easier to develop and maintain.

- **Model:** Represents the data and business logic using Java classes such as *Property*, *User*, and *Inquiry*.
- **View:** Consists of JSP pages that display data to users and collect input.
- **Controller:** Implemented using Java Servlets, which handle user requests, process data, and control the flow between the view and model.

The development process involves designing the database, creating model classes, implementing servlets for different functionalities, and developing JSP pages for user interaction. JDBC is used to connect the application with the MySQL database.

#### 5.2 Module Descriptions

The system is divided into several modules, each responsible for a specific functionality.

##### 5.2.1 Admin Module

This module handles administrator authentication and session management.

- Administrators log in using valid username and password
- Session management is implemented using HttpSession
- Only authenticated admins can access result management features
- Admin credentials are stored securely in the database
- Logout functionality clears the session

### **5.2.2 Result Management Module**

This module allows administrators to add, update, and delete student result records.

- Admins can add new results with details such as roll number, student name, subject, marks, and semester
- Result entries are stored in the database with automatic grade calculation
- Admins can update existing result records
- Admins can delete incorrect or outdated result entries
- All operations maintain data integrity

### **5.2.3 Search Module**

This module provides functionality for students to search their results.

- Students enter their roll number or registration number
- The system performs a search using SQL queries
- Matching results are retrieved from the database
- If no results found, appropriate message is displayed
- Helps students quickly access their academic records

### **5.2.4 Result Detail Module**

This module displays detailed mark sheets to students.

- Shows complete result information including subject-wise marks
- Displays total marks, percentage, and overall grade
- Presents data in a structured and readable format
- Allows students to view semester-wise performance
- Improves transparency and student satisfaction

### **5.2.5 Student Verification Module**

This module ensures that only valid student data is entered.

- Validates roll number format and uniqueness
- Checks for duplicate entries before adding results
- Ensures all required fields are filled
- Prevents invalid or incomplete data submission
- Maintains database accuracy

### **5.2.6 Login and Authentication Module**

This module ensures secure access to the administrative panel.

- Validates admin credentials during login
- Uses session management to maintain admin state
- Prevents unauthorized access to result management features
- Handles login and logout operations securely
- Implements password protection

### **5.2.7 Delete/Update Result Module**

This module allows administrators to modify or remove their property listings.

- Admins can update marks, grades, or other details
- Admins can delete incorrect result entries
- Confirmation prompts prevent accidental deletions
- Maintains audit trail of changes
- Ensures data consistency

## CHAPTER 6

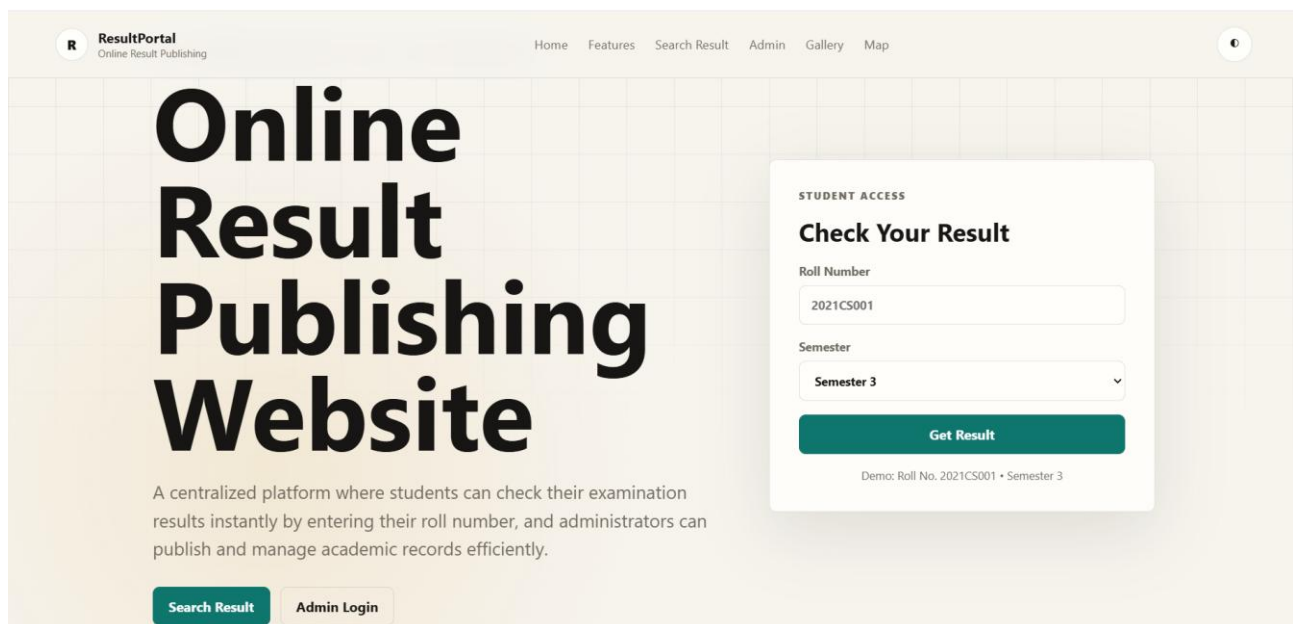
### RESULTS AND DISCUSSION

#### 6.1 System Output Screens

The developed system was tested by executing all major functionalities, and the outputs were verified through different user interactions. The application successfully displays dynamic content retrieved from the database and responds correctly to user inputs.

The main output screens of the system include:

- **Student Homepage (Result Search):**



The screenshot displays the 'ResultPortal' website. The header includes a logo with the letter 'R', the text 'ResultPortal' and 'Online Result Publishing', and navigation links: Home, Features, Search Result, Admin, Gallery, and Map. A user profile icon is in the top right. The main content area features a large heading 'Online Result Publishing Website' and a subtext: 'A centralized platform where students can check their examination results instantly by entering their roll number, and administrators can publish and manage academic records efficiently.' Below this are two buttons: 'Search Result' and 'Admin Login'. On the right, a 'STUDENT ACCESS' box titled 'Check Your Result' contains a 'Roll Number' input field with the value '2021CS001', a 'Semester' dropdown menu set to 'Semester 3', and a 'Get Result' button. At the bottom of this box, it says 'Demo: Roll No. 2021CS001 • Semester 3'.

Displays a search interface where students can enter their roll number to retrieve results.

The page is simple and intuitive, with clear instructions.

- **Admin Login Page:**

Allows authorized administrators to log in using their username and password. Proper validation messages are displayed for incorrect credentials.

The screenshot shows the 'ADMIN MODULE' page of the ResultPortal. On the left, a large heading reads 'Administrator login & result management'. On the right, there is a login form with fields for 'Username' (containing 'admin') and 'Password' (containing 'Enter password'). Below these is a green 'Login to Admin Panel' button. A demo message states 'Demo admin: admin • Password: admin123'. Further down are input fields for 'Roll Number' (2021CS001), 'Student Name' (Student name), 'Class' (B.Tech CSE), and 'Subject Name' (Database Management). At the top, navigation links include Home, Features, Search Result, Admin, Gallery, and Map.

- **Admin Dashboard:**

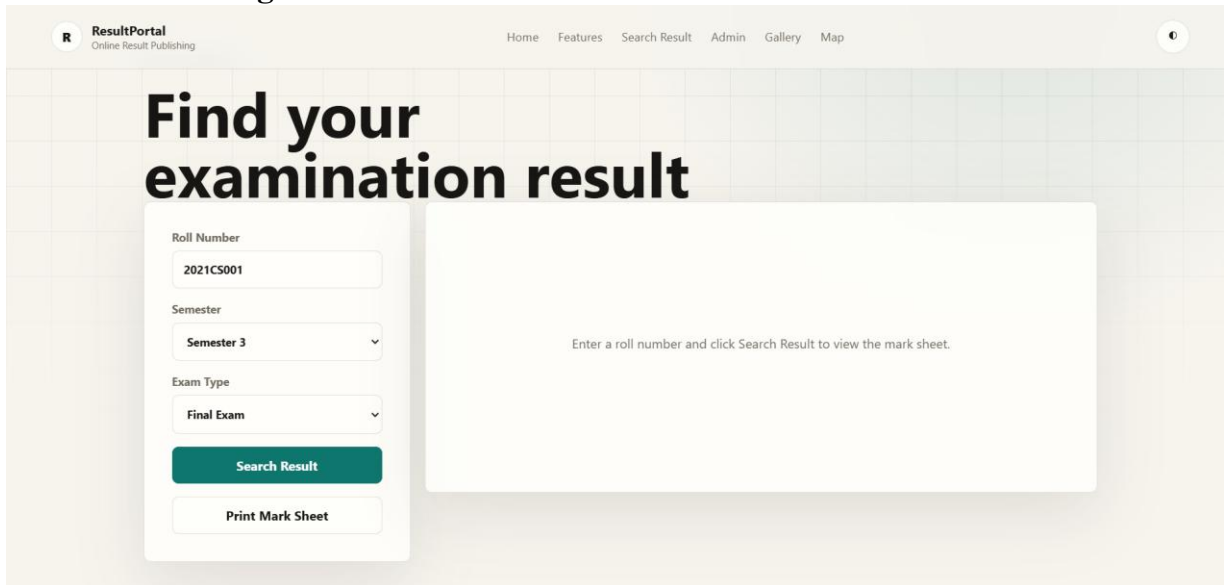
The screenshot shows the 'Admin Dashboard' page of the ResultPortal. It features a form for adding student results. Fields include 'Class' (B.Tech CSE), 'Subject Name' (Database Management), 'Marks Obtained' (85), 'Max Marks' (100), 'Semester' (Semester 3), and 'Exam Type' (Final). A green 'Add Result' button is at the bottom. The 'Roll Number' field is partially visible at the very bottom. The top navigation bar is identical to the login page.

Displays options for adding, updating, viewing, and deleting student results. Provides easy navigation to all administrative functions.

- **Add Results Page:**

Enables administrators to enter student result details including roll number, name, subject, marks, grade, and semester. Includes validation for all input fields.

- **Search Result Page:**



The screenshot shows the 'ResultPortal' website with the title 'Find your examination result'. The navigation bar includes links for Home, Features, Search Result, Admin, Gallery, and Map. The search form on the left contains the following fields and options:

- Roll Number:** Input field with the value '2021CS001'.
- Semester:** Dropdown menu with 'Semester 3' selected.
- Exam Type:** Dropdown menu with 'Final Exam' selected.
- Search Result:** A green button to submit the search.
- Print Mark Sheet:** A button to print the results.

To the right of the form, a message states: 'Enter a roll number and click Search Result to view the mark sheet.'

Shows the student's complete mark sheet after successful roll number search. Displays subject-wise marks, total, percentage, and grade in a formatted table.

- **Result Details Page:**

Presents detailed information including student name, roll number, class, semester, exam type, and individual subject performance.

- **Delete Confirmation Page:**

- Prompts administrators to confirm before deleting a result entry, preventing accidental data loss.
- These output screens demonstrate that the system is functioning correctly and providing the intended features with accuracy and clarity.

- **Update Result Page:**

Allows administrators to modify existing result entries. Pre-fills current data and allows editing.



These output screens demonstrate that the system is functioning correctly and providing the intended features.

## **6.2 Performance Evaluation**

The system was evaluated based on responsiveness, accuracy, and usability.

- **Response Time:**

The system responds quickly to user requests such as searching results, adding new entries, and displaying mark sheets. Average response time is under 2 seconds for most operations.

- **Data Accuracy:**

All data retrieved from the database is displayed correctly without inconsistencies. Calculations for total marks, percentage, and grades are accurate.

- **Security:**

The system implements session-based authentication for administrators, preventing unauthorized access. Password fields are protected, and SQL injection vulnerabilities are minimized through prepared statements.

- **Usability:**

The user interface is simple, clean, and easy to navigate. Students can search results without confusion, and administrators can manage data efficiently.

- **Efficiency:**

The use of JDBC and optimized SQL queries ensures efficient data handling and retrieval. The database is properly indexed for faster searches.

Overall, the system performs well for small to medium-scale educational institutions and provides a smooth, reliable user experience.

## **6.3 System Testing**

The system was tested using different scenarios to ensure reliability and

correctness.

- **Functional Testing:**

All features such as admin login, result addition, student search, result display, update, and delete were tested and found to be working as expected.

- **Authentication Testing:**

Admin login and session management were tested to ensure secure access. Unauthorized users cannot access administrative functions.

- **Validation Testing:**

Input fields were tested for correct validation, ensuring that invalid, empty, or incorrect data formats are not accepted. Roll number uniqueness is enforced.

- **Database Testing:**

Data insertion, retrieval, update, and deletion operations were verified to ensure proper interaction with the database. Foreign key constraints and data integrity were validated.

- **Error Handling:**

The system handles errors gracefully and displays appropriate messages to the user. For example, "No results found" when a roll number does not exist.

The testing results confirm that the system is stable, reliable, secure, and ready for deployment in an educational environment.

## **CHAPTER 7**

### **CONCLUSION AND FUTURE WORK**

#### **7.1 Conclusion**

The online result publishing system was successfully designed and developed to simplify the process of publishing, viewing, and managing academic examination results. The application provides a user-friendly interface that allows students to check their results quickly by entering their roll number, and enables administrators to publish and manage result records efficiently.

The system effectively integrates frontend and backend technologies using Java Servlets, JSP, and MySQL. It follows a structured MVC architecture that ensures clear separation of components, making the application easy to understand, maintain, and extend. Key features such as administrator authentication, result management, automated grade calculation, student result search, and secure database interaction have been implemented successfully.

The project demonstrates the practical application of web development concepts and database management in the education sector. It provides a reliable, transparent, and efficient platform for handling examination results, eliminating the need for manual result distribution and reducing delays. Overall, the system meets its objectives and performs efficiently for small to medium-scale educational institutions.

## 7.1 Future Enhancements

Although the system meets the current requirements, it can be further improved by adding more advanced features in the future:

- **Advanced Search Filters:**  
Adding filters such as semester-wise results, subject-wise performance, class-wise results, and date range for more precise searches.
- **Result Analytics Dashboard:**  
Displaying performance graphs, charts, subject-wise analysis, pass/fail statistics, and semester comparison for better insights.
- **PDF and Print Support:**  
Allowing students to download or print their mark sheets in PDF format for official use and record keeping.
- **Mobile Application:**  
Developing a mobile version of the system (Android/iOS) for improved accessibility and convenience.
- **Bulk Result Upload:**  
Providing an option for administrators to upload results in bulk using Excel or CSV files, reducing manual data entry time.
- **Student Portal:**  
Creating a dedicated student login where they can view all semester results, track academic progress, and download certificates.
- **Multi-level Access Control:**  
Implementing different roles such as Super Admin, Faculty, and Data Entry Operator with specific permissions.
- **Result verification System:**  
Adding QR code or unique verification code on mark sheets for authenticity verification by external organizations.
- **Admin Panel:**  
Providing an admin interface to manage users, properties, and inquiries
- **Email Notifications:**  
Sending automatic notifications to students when results are published, including a link to check results online.
- **Improved Security:**  
Enhancing security with password encryption (hashing), two-factor authentication, and CAPTCHA on login pages.

These enhancements can make the system more scalable, interactive, and suitable for real-world deployment on a larger scale.

## REFERENCES

- [1] Oracle Corporation, *Java Servlet Technology Documentation*.  
Available: <https://docs.oracle.com/javaee/>
- [2] Oracle Corporation, *JSP (Java Server Pages) Documentation*.  
Available: <https://docs.oracle.com/javaee/>
- [3] Oracle Corporation, *JDBC API Documentation*.  
Available: <https://docs.oracle.com/javase/>
- [4] Apache Software Foundation, *Apache Tomcat Documentation*.  
Available: <https://tomcat.apache.org/>
- [5] MySQL Documentation Team, *MySQL Reference Manual*.  
Available: <https://dev.mysql.com/doc/>
- [6] MDN Web Docs, *HTML, CSS, and Web Development Guides*.  
Available: <https://developer.mozilla.org/>
- [7] W3Schools, *Web Development Tutorials*.  
Available: <https://www.w3schools.com/>
- [8] GeeksforGeeks, *Java and Web Development Articles*.  
Available: <https://www.geeksforgeeks.org/>
- [9] K. Sierra and B. Bates, *Head First Servlets and JSP*, O'Reilly Media, 2nd Edition, 2008.
- [10] E. Balagurusamy, *Programming with Java*, McGraw Hill Education, 6th Edition.

## APPENDICES

### Appendix-1: Sample Database Tables

#### Admin Table

| Field Name | Data Type | Description     |
|------------|-----------|-----------------|
| admin_id   | INT (PK)  | Unique admin ID |
| username   | VARCHAR   | Admin username  |
| password   | VARCHAR   | Admin password  |
| email      | VARCHAR   | Admin email     |

#### Students Table

| Field Name  | Data Type        | Description         |
|-------------|------------------|---------------------|
| student_id  | INT (PK)         | Unique student ID   |
| roll_number | VARCHAR (UNIQUE) | Student roll number |
| name        | VARCHAR          | Student name        |
| class       | VARCHAR          | Class/Year          |
| section     | VARCHAR          | Section             |

#### Results Table

| Field Name     | Data Type | Description           |
|----------------|-----------|-----------------------|
| result_id      | INT (PK)  | Unique result ID      |
| student_id     | INT (FK)  | Reference to student  |
| roll_number    | VARCHAR   | Student roll number   |
| subject_name   | VARCHAR   | Subject name          |
| marks_obtained | INT       | Marks scored          |
| max_marks      | INT       | Maximum marks         |
| grade          | VARCHAR   | Grade (A, B, C, etc.) |
| semester       | VARCHAR   | Semester/Term         |
| exam_type      | VARCHAR   | Exam type (Mid/Final) |

## **Appendix-2: Sample SQL Queries**

-- Insert Admin

```
INSERT INTO admin (username, password, email)
VALUES ('admin', 'admin123', 'admin@school.com');
```

-- Insert Student

```
INSERT INTO students (roll_number, name, class, section)
VALUES ('2021CS001', 'Rahul Sharma', 'B.Tech CSE', 'A');
```

-- Insert Result

```
INSERT INTO results (student_id, roll_number, subject_name, marks_obtained, max_marks, grade,
semester, exam_type)
VALUES (1, '2021CS001', 'Database Management', 85, 100, 'A', 'Semester 4', 'Final');
```

-- Search Result by Roll Number

```
SELECT * FROM results
WHERE roll_number = '2021CS001';
```

-- Get Student Result with Details

```
SELECT s.name, s.roll_number, s.class, r.subject_name, r.marks_obtained, r.grade
FROM students s
JOIN results r ON s.student_id = r.student_id
WHERE s.roll_number = '2021CS001';
```

-- Calculate Total and Percentage

```
SELECT roll_number,
       SUM(marks_obtained) AS total_marks,
       SUM(max_marks) AS total_max_marks,
       (SUM(marks_obtained) / SUM(max_marks)) * 100 AS percentage
FROM results
WHERE roll_number = '2021CS001'
GROUP BY roll_number;
```

-- Update Result

```
UPDATE results
SET marks_obtained = 90, grade = 'A+'
WHERE result_id = 1;
```

-- Delete Result

```
DELETE FROM results
WHERE result_id = 1;
```



### Appendix-3: Key Servlet Example

```
@WebServlet("/searchResult")
public class SearchResultServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String rollNumber = request.getParameter("rollNumber");

        List<Result> resultList = new ArrayList<>();
        Connection con = DBConnection.getConnection();

        try {
            String query = "SELECT s.name, s.roll_number, s.class, " +
                "r.subject_name, r.marks_obtained, r.max_marks, r.grade " +
                "FROM students s JOIN results r ON s.student_id = r.student_id " +
                "WHERE s.roll_number = ?";

            PreparedStatement pst = con.prepareStatement(query);
            pst.setString(1, rollNumber);
            ResultSet rs = pst.executeQuery();

            while (rs.next()) {
                Result r = new Result();
                r.setStudentName(rs.getString("name"));
                r.setRollNumber(rs.getString("roll_number"));
                r.setSubjectName(rs.getString("subject_name"));
                r.setMarksObtained(rs.getInt("marks_obtained"));
                r.setMaxMarks(rs.getInt("max_marks"));
                r.setGrade(rs.getString("grade"));
                resultList.add(r);
            }

            if (resultList.isEmpty()) {
                request.setAttribute("message", "No results found for this roll number");
            }

        } catch (Exception e) {
            e.printStackTrace();
        }

        request.setAttribute("resultList", resultList);
        request.getRequestDispatcher("viewResult.jsp").forward(request, response);
    }
}
```

# PSEUDOCODE

The following pseudocode represents the core functionalities of the online result publishing system.

## 1. Admin Login

```
BEGIN
  INPUT username, password
  CONNECT to database
  FETCH admin where username and password match
  IF admin exists THEN
    CREATE session
    STORE admin details in session
    REDIRECT to admin dashboard
  ELSE
    DISPLAY "Invalid credentials"
  ENDIF
  CLOSE database connection
END
```

## 2. Add Result

```
BEGIN
  CHECK if admin is logged in
  INPUT roll_number, student_name, subject_name,
marks_obtained,max_marks, semester, exam_type
  CONNECT to database

  CALCULATE grade based on marks_obtained:
    IF marks >= 90 THEN grade = "A+"
    ELSE IF marks >= 80 THEN grade = "A"
    ELSE IF marks >= 70 THEN grade = "B+"
    ELSE IF marks >= 60 THEN grade = "B"
    ELSE IF marks >= 50 THEN grade = "C"
    ELSE grade = "F"

  CHECK if student exists in students table
  IF NOT exists THEN
    INSERT student details into students table
  ENDIF

  INSERT result details into results table
  DISPLAY "Result added successfully"
  CLOSE database connection
END
```

### 3. Search Result by Roll Number

```
BEGIN
  INPUT roll_number
  CONNECT to database

  EXECUTE query:
    SELECT s.name, s.roll_number, s.class, r.subject_name,
           r.marks_obtained, r.max_marks, r.grade, r.semester
    FROM students s
    JOIN results r ON s.student_id = r.student_id
    WHERE s.roll_number = roll_number

  IF results found THEN
    STORE results in list
    CALCULATE total_marks = SUM(marks_obtained)
    CALCULATE total_max_marks = SUM(max_marks)
    CALCULATE percentage = (total_marks / total_max_marks) * 100
    DISPLAY student details and subject-wise results
    DISPLAY total, percentage, overall grade
  ELSE
    DISPLAY "No results found for this roll number"
  ENDIF

  CLOSE database connection
END
```

### 4. View All Results (Admin)

```
BEGIN
  CHECK if admin is logged in
  CONNECT to database

  FETCH all results ordered by roll_number and semester
  STORE results in list
  SEND list to JSP page
  DISPLAY results in tabular format

  CLOSE database connection
END
```

### 5. Update Result

```
BEGIN
  CHECK if admin is logged in
  GET result_id from request
  CONNECT to database

  INPUT new marks_obtained
```

```
CALCULATE new grade based on updated marks

UPDATE results table:
    SET marks_obtained = new_marks, grade = new_grade
    WHERE result_id = result_id

DISPLAY "Result updated successfully"
CLOSE database connection
END
```

## 6. Delete Result

```
BEGIN
    CHECK if admin is logged in
    GET result_id from request
    CONNECT to database

    DELETE FROM results WHERE result_id = result_id

    DISPLAY "Result deleted successfully"
    CLOSE database connection
END
```

## 7. Calculate Overall Grade

```
BEGIN
    INPUT list of subject grades
    INITIALIZE grade_points = 0
    INITIALIZE total_subjects = length of list

    FOR each grade in list DO
        IF grade = "A+" THEN add 10
        ELSE IF grade = "A" THEN add 9
        ELSE IF grade = "B+" THEN add 8
        ELSE IF grade = "B" THEN add 7
        ELSE IF grade = "C" THEN add 6
        ELSE add 0
    ENDFOR

    CALCULATE average = grade_points / total_subjects

    IF average >= 9 THEN overall_grade = "A+"
    ELSE IF average >= 8 THEN overall_grade = "A"
    ELSE IF average >= 7 THEN overall_grade = "B+"
    ELSE IF average >= 6 THEN overall_grade = "B"
    ELSE overall_grade = "C"

    RETURN overall_grade
END
```

## 8. Admin Layout

```
BEGIN  
  GET current session  
  INVALIDATE session  
  REDIRECT to login page  
END
```